

Part C: Direct AS1–AS4 Peering and Delay Comparison

COL7524 – Advanced Computer Networks

Objective. Starting from the Part B topology, implement a policy so that AS1 prefers to send traffic through AS2 and AS3, even though a faster direct AS1–AS4 link is available. The policy is implemented using two mechanisms: a static route and an interface-metric adjustment in ns-3's global routing.

1. Part C Topology

Part C uses the Part B topology, which has two paths between AS1 and AS4:

Three-hop path	AS1 → AS2 → AS3 → AS4
Direct path	AS1 → AS4
Direct-link configuration	10 Mbps, 2 ms

2. Mechanism 1 - Static Route

The static mechanism explicitly changes the forwarding decision at AS1. For the AS4 server address 192.168.3.2, AS1 is configured with the next hop 192.168.1.2, which is the AS2-side address of the AS1–AS2 link.

Forward: AS1 → AS2 → AS3 → AS4

In the simplified topology, an additional static forwarding entry was added at AS2 to ensure packets destined for 192.168.3.2 continue through AS3 and are not affected by a competing global route back toward AS1.

The return traffic did not follow the static policy of AS1 and therefore used the direct link AS4 → AS1.

Return: AS4 → AS1

The forward flow delivered all 8 packets without loss. Its delaySum was 9.44533×10^7 ns, giving an average delay of approximately 11.807 ms. The reverse flow also delivered all 8 packets, with delaySum 1.94688×10^7 ns and an average delay of 2.4336 ms.

3. Mechanism 2 - Interface Metric / Cost

The metric mechanism does not specify a next hop. Instead, interface costs are changed before global routing tables are populated so that the desired path has the lower routing cost. On AS1, the AS1→AS2 interface is assigned metric 1, and the direct AS1→AS4 interface metric 10. Symmetric values are applied at AS4 so that the return direction also prefers AS3 rather than the direct AS4–AS1 link.

Node	Interface/link	Metric
AS1	AS1 → AS2	1
AS1	AS1 → AS4	10
AS4	AS4 → AS3	1
AS4	AS4 → AS1	10

After the metrics are set, `Ipv4GlobalRoutingHelper::PopulateRoutingTables()` recomputes the routes using the modified interface costs.

Forward: AS1 → AS2 → AS3 → AS4

Return: AS4 → AS3 → AS2 → AS1

Both flows delivered all 8 packets with zero loss, and both reported an average delay of approximately 11.807 ms.

4. Comparison of the Two Mechanisms and Results

Mechanism	Policy method	Direction	Observed path	Average delay
Static route	AS1 explicitly uses AS2 as the next hop; AS2 is then forwarded toward AS3.	Forward	AS1 → AS2 → AS3 → AS4	11.807 ms
Static route	The policy applies to AS1's outbound forwarding; no return-path policy is imposed.	Return	AS4 → AS1	2.434 ms

Interface metric	A higher cost for the direct interfaces causes global routing to select the lower-cost chain.	Forward / Return	AS1 → AS2 → AS3 → AS4 / AS4 → AS3 → AS2 → AS1	11.807 ms
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Both mechanisms enforce the same outbound routing preference but in different ways. The static-route mechanism explicitly states the next hop at AS1 (and the forwarding entry at AS2). The interface-metric mechanism modifies the routing costs so that global routing chooses the preferred path. In the static experiment, the forward path is AS1 → AS2 → AS3 → AS4, and the return path is AS4 → AS1. In the metric experiment, the forward path is AS1→AS2→AS3→AS4, and the return path is AS4→AS3→AS2→AS1. In both experiments, there is no loss in all 8 packets. The forward delay is about 11.807 ms.

5. Conclusion

The experiments show that routing policy can cause a network to prefer a slower multi-hop path over an available direct one. Both static routing and interface-metric adjustment work well to achieve the required preference, but they differ in how that preference is enforced.